

Multi-Directional Bicycle Blind Spot & Collision Alert System

1. Abstract

Road accidents involving bicycle and rickshaw riders are becoming a serious problem, especially in busy city areas where vehicles often come from the rear and side without being noticed. Because of blind spots and limited visibility, riders sometimes fail to see approaching vehicles, which increases the chance of accidents. This project presents a **Multi-Directional Bicycle Blind Spot & Collision Alert System**, which is a low-cost and real-time safety solution designed to improve rider awareness.

This system uses three ultrasonic sensors positioned on the left, right and back sides of a bicycle to monitor nearby vehicles continuously. An Arduino Uno processes the distance data and estimates the speed of nearby objects by analyzing changes in distance over time. Based on the calculated risk level, the system provides real-time warnings through an LCD display, buzzer alerts, and a critical red LED indicator.

The proposed system categorizes road situations into Safe, Warning, High Risk and Critical states. It prioritizes dangerous directions and provides immediate alerts only when actual collision risks are detected. Experimental testing demonstrated that the system can successfully detect approaching vehicles, estimate danger levels, and provide timely warnings to the rider.

This project offers a simple, affordable, and effective approach to improving road safety for cyclists, delivery riders and rickshaw pullers. Moreover, the system provides a foundation for future enhancements such as GPS integration, mobile application connectivity, and AI-based risk prediction.

Keywords: Arduino Uno, Ultrasonic Sensor, Blind Spot Detection, Collision Alert System, Bicycle Safety, Real-Time Monitoring.

2. Introduction

2.1 Background

In Bangladesh, bicycles and rickshaws are very commonly used because they are affordable and easy to use. Many students, delivery workers and rickshaw pullers

depend on them for daily travel. However, unlike modern automobiles, bicycles and rickshaws lack advanced safety technologies that can warn riders about approaching vehicles. Unlike automobiles, bicycles lack any system to simultaneously monitor left, right, and back directions, leaving riders completely unaware of multi-directional threats.

Riders frequently fail to notice vehicles approaching from rear and side directions due to limited visibility. In busy traffic environments, riders often become unaware of fast-moving vehicles, which may lead to serious collisions and injuries.

Recent accident reports indicate that many cyclists lose their lives due to rear-end or side-impact collisions. Most existing safety solutions are either designed primarily for automobiles, making them unsuitable for widespread use among bicycle and rickshaw users.

2.2 Motivation

The motivation behind this project is to develop a simple and affordable road safety system that can provide real-time awareness of nearby vehicles. The system aims to assist riders by continuously monitoring blind spots and alerting them whenever a potentially dangerous situation is detected.

The project was specifically designed considering the transportation conditions of Bangladesh, where:

- i. Bicycle usage is increasing.
- ii. Delivery services depend heavily on bicycles.
- iii. Rickshaws remain one of the most common vehicles.
- iv. Affordable safety solutions are rarely available.

By integrating low-cost sensors and an Arduino microcontroller, the project seeks to improve rider safety without significantly increasing the cost of the vehicle.

2.3 Problem Statement

Cyclists and rickshaw riders often face difficulty detecting vehicles approaching from the left, right, and back sides. Due to limited visibility and blind spots, riders may fail to react appropriately, increasing the probability of accidents.

The major problems identified are:

- i. Cannot detect vehicles in blind spots.
- ii. No cheap smart safety system for bicycles and rickshaws.
- iii. Higher chance of accidents because riders notice danger late.

iv. No real-time system to check danger while riding.

Because of these problems, a simple collision alert system is needed that can watch nearby traffic and warn the rider quickly when danger appears.

2.4 Objectives

The primary objectives of this project are:

General Objective

To develop a smart collision alert system that improves the safety of bicycle and rickshaw riders through real-time monitoring and warning mechanisms.

Specific Objectives

1. Detects vehicles approaching from the left, right, and back directions.
2. Measure the distance of nearby objects using ultrasonic sensors.
3. Estimate the speed of approaching vehicles using distance variation analysis.
4. Classify situations into Safe, Warning, High Risk, and Critical states.
5. Display real-time information using an LCD screen.
6. Generate buzzer alerts based on risk levels.
7. Activate a red safety LED during critical situations.
8. Reduce false alerts through averaging and filtering techniques.

2.5 Scope of the Project

The project focuses on providing real-time collision warnings for bicycle and rickshaw riders using ultrasonic sensing technology. The system operates within the detection range of HC-SR04 ultrasonic sensors and provides localized warnings based on detected object distance and estimated approach speed.

The scope includes:

- i. Multi-directional obstacle detection.
- ii. Distance measurement.
- iii. Risk assessment.
- iv. LCD-based information display.
- v. Buzzer and LED alerts.

The scope does not include:

- i. Vehicle identification.
- ii. GPS navigation.
- iii. Wireless communication.
- iv. Autonomous vehicle control.

These features are considered potential future enhancements.

2.6 Project Overview

The proposed system consists of three ultrasonic sensors mounted on the left, right, and rear sides of a bicycle. The sensors continuously measure object distances and transmit data to an Arduino Uno microcontroller.

The Arduino processes the sensor readings, determines the nearest object, estimates its approach speed, and evaluates the risk level. Depending on the calculated danger level, the system generates appropriate alerts:

- i. Safe: No alert.
- ii. Warning: Slow buzzer beep.
- iii. High Risk: Fast buzzer beep.

Critical: Continuous buzzer and blinking red LED.

At the same time, the LCD display presents the current system status along with the detected direction and distance information.

The overall workflow enables riders to receive timely warnings and react before a collision occurs. Also, the LCD is capable of simultaneously displaying multi-directional object distances along with their corresponding threat levels in real time.

3. Design

3.1 Problem Formulation

Bicycle and rickshaw riders frequently encounter dangerous situations caused by vehicles approaching from blind spots. Since riders cannot continuously monitor the rear and side directions, collision risks increase significantly in busy traffic environments.

The problem can be formulated as follows:

Design a low-cost real-time safety system capable of detecting vehicles approaching from the left, right, and rear directions, analyzing the collision risk, and providing immediate warnings to the rider before an accident occurs.

3.2 Design Method

The project follows a sensor-based embedded system design approach.

The overall operation consists of the following steps:

Step 1: Object Detection

Three HC-SR04 ultrasonic sensors continuously measure the distance of nearby vehicles.

Sensor Placement:

1. Left Sensor → Left side traffic monitoring
2. Right Sensor → Right side traffic monitoring
3. Back Sensor → Back traffic monitoring

Step 2: Distance Measurement

Each ultrasonic sensor transmits an ultrasonic pulse and measures the echo return time.

The distance is calculated using:

Where:

- i. Speed of Sound = 0.034 cm/ μ s
- ii. Time = Echo duration
- iii. Distance (cm) = (Echo Time $\times$ 0.034) / 2

To improve accuracy, the system takes five readings and calculates the average value.

Step 3: Speed Approximation

The project estimates approaching speed using changes in distance over time.

The Arduino stores:

- i. Previous distance
- ii. Current distance
- iii. Previous time
- iv. Current time

Approaching speed is calculated as:

- i. Speed (cm/s) = (Previous Distance – Current Distance) / Time Elapsed (s)

A rapidly decreasing distance indicates a fast-approaching vehicle.

Step 4: Risk Evaluation

The system classifies danger into four categories:

Table 3.1: Risk Evaluation

Risk level	Condition
Safe	$d \geq 40 \text{ cm}$
Warning	$d < 40 \text{ cm}$
High Risk	$d < 25 \text{ cm}$, OR ($d < 30 \text{ cm}$ AND speed $> 5 \text{ cm/s}$)
Critical	$d < 10 \text{ cm}$, OR ($d < 20 \text{ cm}$ AND speed $> 10 \text{ cm/s}$)

Step 5: Alert Generation

The rider receives alerts through:

- i. LCD Display
- ii. Buzzer
- iii. Red LED

The alert intensity increases according to risk level.

3.3 Hardware Components

The following hardware components were used in the implementation.

Table 3.2: Required Components

Component	Quality	Purpose
Arduino UNO	1	Main controller
HC-SR04 Ultrasonic Sensor	3	Distance measurement
LCD 16×2 with I2C Module	1	Display information
Active Buzzer	1	Audible alert
Red LED	1	Critical danger indicator
Breadboard	1	Circuit assembly
Jumper Wires	Multiple	Connections
Power Bank / Battery	1	Power source

3.4 Circuit Diagram

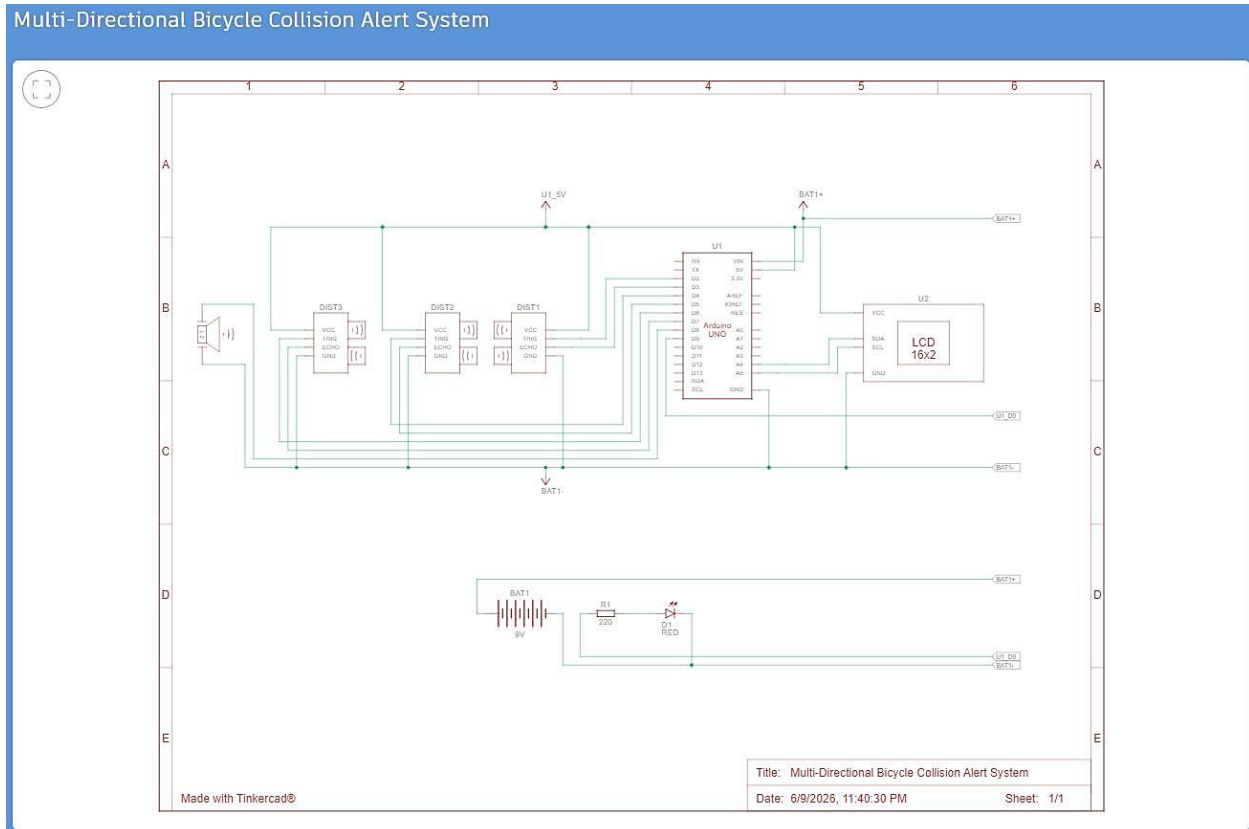


Figure 3.1 Complete circuit implementation of the proposed system.

Circuit Description:

The Arduino Uno acts as the central processing unit.

Three HC-SR04 sensors are connected to separate digital pins:

Table 3.3: Sensor Connection

Sensor	Trigger Pin	Echo Pin
Left sensor	D2	D3
Right sensor	D4	D5
Back sensor	D6	D7

The output devices are connected as follows:

Table 3.4: LCD, Led and Buzzer Connection

Device	Arduino Pin
Buzzer	D8
Red LED	D9
LCD SDA	A4
LCD SCL	A5

The LCD uses the I2C communication protocol, reducing wiring complexity and minimizing pin usage.

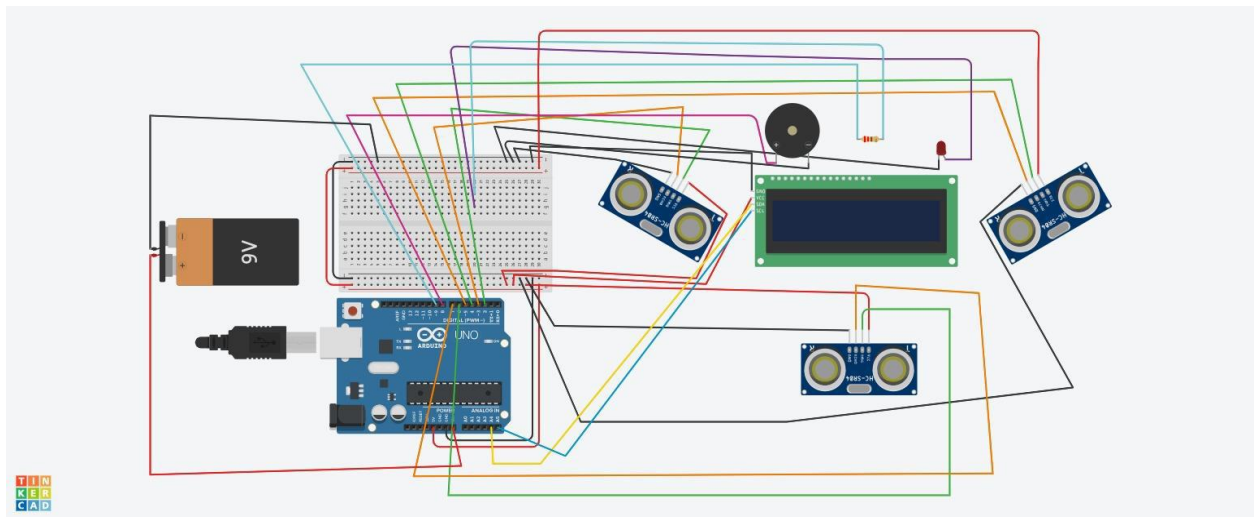


Figure 3.2: Tinkercard simulated circuit implementation of the proposed system.

3.5 Working Principle

The system operates continuously after power is supplied. The LCD continuously updates and displays left, right, and back sensor readings simultaneously along with risk level classification.

When multiple directions are at risk simultaneously, the LCD displays all affected directions together. For example, if both left and right sensors detect danger: HIGH RISK / L:22 R:29

i. Safe State

When no vehicle is detected within the danger range:

LCD Output:

ALL SAFE

Buzzer: OFF

LED: OFF

ii. Warning State

When a vehicle enters the warning zone:

LCD Output:

WARNING

R:35

or

WARNING

B:38

Buzzer:

i. Slow beep

LED:

ii. OFF

iii. High Risk State

When a vehicle approaches more closely:

LCD Output:

HIGH RISK

L:22

Buzzer:

i. Fast double beep

LED:

ii. OFF

iv. Critical State

When collision risk becomes very high:

LCD Output:

CRITICAL

B:8

Buzzer:

- i. Continuous sound

LED:

- ii. Fast blinking red light

This state provides the strongest warning to the rider.

3.6 False Alert Reduction Method

Traffic environments contain noise and sudden measurement fluctuations.

To improve reliability, the system:

Averaging Technique

Instead of using a single reading:

- i. Five consecutive measurements are taken.
- ii. Their average value is calculated.

This reduces sensor noise.

Risk Filtering

The system combines:

- i. Distance
- ii. Estimated speed

before generating alerts.

This prevents unnecessary warnings caused by temporary obstacles.

3.7 Algorithm

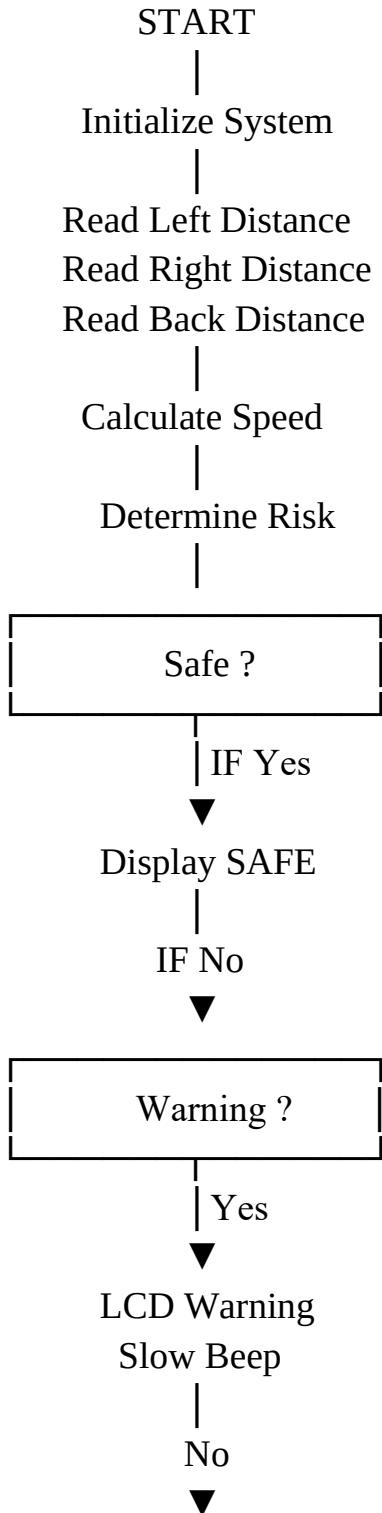
The algorithm of the proposed system is summarized below.

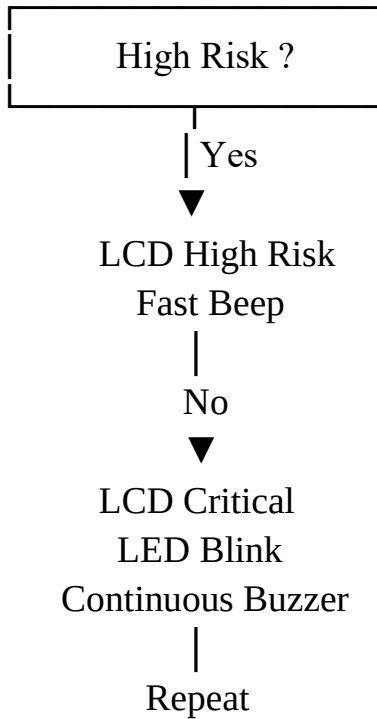
Algorithm Steps

1. Start system.
2. Then initialize LCD, sensors, buzzer and LED.
3. Read left sensor distance.
4. Then read the distance from the right sensor.
5. Next, read the distance from the rear sensor.
6. Then determine the nearest object distance.
7. Calculate the speed approximation.
8. Next, determine risk level.
9. Display status on LCD.
10. Then generate the appropriate buzzer alerts.
11. If a critical condition is detected, turn on the red LED.

12.Finally, repeat the process continuously.

3.8 Flowchart





3.9 Code

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Sensors

const int trigL = 2;

const int echoL = 3;

const int trigR = 4;

const int echoR = 5;

const int trigB = 6;

const int echoB = 7;

// Outputs

const int buzzer = 8;

const int redLED = 9;

//Speed tracking

float prevDist = 0;
```

```

float speed = 0;

unsigned long prevTime = 0;

// LED blink

bool ledState = false;

unsigned long lastBlink = 0;

// Distance

long getDistance(int trig, int echo){

    long sum = 0;

    for (int i = 0; i < 5; i++) {

        digitalWrite(trig, LOW);

        delayMicroseconds(2);

        digitalWrite(trig, HIGH);

        delayMicroseconds(10);

        digitalWrite(trig, LOW);

        long duration = pulseIn(echo, HIGH);

        long dist = duration * 0.034 / 2;

        sum += dist;

        delay(5); }

    return sum / 5;}

// RISK and SPEED

int getRisk(long d, float s){

    if (d < 10) return 3;

    else if (d < 20 && s > 10) return 3;

    else if(d < 25) return 2;

    else if (d < 30 && s > 5) return 2;

    else if (d < 40) return 1;

    return 0;}

// Setup

void setup(){

```

```
lcd.init();  
lcd.backlight();  
  
pinMode(trigL, OUTPUT);  
pinMode(echoL, INPUT);  
pinMode(trigR, OUTPUT);  
pinMode(echoR, INPUT);  
pinMode(trigB, OUTPUT);  
pinMode(echoB, INPUT);  
pinMode(buzzer, OUTPUT);  
pinMode(redLED, OUTPUT);  
lcd.setCursor(0, 0);  
lcd.print("System Ready");  
delay(1500);  
lcd.clear();  
prevTime = millis();}  
  
// Loop  
void loop(){  
    long dL = getDistance(trigL, echoL);  
    long dR = getDistance(trigR, echoR);  
    long dB = getDistance(trigB, echoB);  
    long currDist = min(dL, min(dR, dB));  
    // SPEED CALCULATION  
    unsigned long now = millis();  
    float dt = (now - prevTime) / 1000.0;  
    if (dt > 0){  
        speed = (prevDist - currDist) / dt;  
        if (speed < 0) speed = 0; }  
    prevDist = currDist;
```

```

prevTime = now;

int rL = getRisk(dL, speed);
int rR = getRisk(dR, speed);
int rB = getRisk(dB, speed);

bool hasCritical = (rL == 3 || rR == 3 || rB == 3);
bool hasHigh    = (rL == 2 || rR == 2 || rB == 2);
bool hasWarning = (rL == 1 || rR == 1 || rB == 1);

// LCD
lcd.clear();

if (hasCritical)
    lcd.print("CRITICAL");
else if (hasHigh)
    lcd.print("HIGH RISK");
else if (hasWarning)
    lcd.print("WARNING");
else
    lcd.print("ALL SAFE");

lcd.setCursor(0, 1);

int activeCount = 0;

if (rL >= 2) activeCount++;
if (rR >= 2) activeCount++;
if (rB >= 2) activeCount++;

if (activeCount == 3){
    if (rR >= 2) { lcd.print("R"); lcd.print(dR); lcd.print(" "); }
    if (rL >= 2) { lcd.print("L"); lcd.print(dL); lcd.print(" "); }
    if (rB >= 2) { lcd.print("B"); lcd.print(dB); lcd.print(" "); } }
else {
    if (rL >= 2) { lcd.print("L:"); lcd.print(dL); lcd.print(" "); }
    if (rR >= 2) { lcd.print("R:"); lcd.print(dR); lcd.print(" "); }

```

```

if (rB >= 2) { lcd.print("B:"); lcd.print(dB); lcd.print(" "); }
if (!hasCritical && !hasHigh) {
  if (rL == 1) { lcd.print("L:"); lcd.print(dL); lcd.print(" "); }
  if (rR == 1) { lcd.print("R:"); lcd.print(dR); lcd.print(" "); }
  if (rB == 1) { lcd.print("B:"); lcd.print(dB); lcd.print(" "); } }
}

```

```
// BUZZER + LED
```

```

if (!hasWarning && !hasHigh && !hasCritical) {
  digitalWrite(buzzer, LOW);
  digitalWrite(redLED, LOW); }
else if (hasWarning && !hasHigh && !hasCritical) {
  digitalWrite(buzzer, HIGH);
  delay(100);
  digitalWrite(buzzer, LOW);
  digitalWrite(redLED, LOW); }
else if (hasHigh && !hasCritical) {
  digitalWrite(redLED, LOW);
  for (int i = 0; i < 2; i++) {
    digitalWrite(buzzer, HIGH);
    delay(150);
    digitalWrite(buzzer, LOW);
    delay(100); } }
else if (hasCritical) {
  digitalWrite(buzzer, HIGH);
  if (millis() - lastBlink >= 200) {
    lastBlink = millis();
    ledState = !ledState;
    digitalWrite(redLED, ledState); } }
delay(150);}

```

3.10 Code Description

The software was developed using Arduino IDE in C/C++ language.

The program contains the following major functions:

getDistance()

This function measures the distance using an HC-SR04 sensor and returns the average of five readings.

Purpose:

- i. Increase accuracy
- ii. Reduce measurement noise

getRisk()

This function determines the danger level based on:

- i. Distance
- ii. Estimated speed

Risk Levels:

Table 3.5: Risk Evaluation

Return Value	Meaning
0	Safe
1	Warning
2	High risk
3	Critical

setup()

The setup function:

- i. Initializes LCD
- ii. Configures sensor pins
- iii. Configures buzzer and LED

iv. Displays "System Ready"

loop()

The main loop continuously:

1. Read all sensors.
2. Calculates speed.
3. Evaluates risk.
4. Updates LCD.
5. Controls buzzer.
6. Controls LED.

The process repeats indefinitely to provide real-time monitoring.

4. Implementation

4.1 Introduction

This chapter describes the practical implementation of the proposed Multi-Directional Smart Bicycle Blind Spot & Collision Alert System. It includes hardware assembly, software implementation, data collection procedures, testing methods, and experimental results obtained during prototype development.

The system was implemented using Arduino Uno, three HC-SR04 ultrasonic sensors, a 16×2 I2C LCD display, a buzzer, and a red LED. The prototype was mounted on a toy bicycle model to simulate real-world operation.

4.2 Hardware Implementation

The hardware components were connected according to the circuit design discussed in Chapter 3.

Sensor Placement

The three ultrasonic sensors were positioned to cover all major blind spot areas:

Table 4.1: Sensor Placement

Sensor	Position
Left Sensor	Left side of bicycle
Right Sensor	Right side of bicycle

Back Sensor	Back side of bicycle
-------------	----------------------

This arrangement allows the system to monitor approaching vehicles from multiple directions simultaneously.

Output Device Placement

Table 4.2: Output Device Placement

Component	Placement
LCD Display	Front handle area
Buzzer	Near handle/front section
Red LED	Front visible area
Arduino Uno	Below bicycle frame

The LCD acts as the primary output device while the buzzer and LED provide immediate alerts.

4.3 Software Implementation

The software was developed using Arduino IDE.

Programming Language:

- i. Embedded C/C++

Main software tasks:

1. Sensor data acquisition
2. Distance averaging
3. Speed estimation
4. Risk classification
5. LCD information display
6. Buzzer control
7. LED control

The Arduino continuously processes data and updates the outputs in real time.

4.4 Data Collection Procedure

To evaluate system performance, multiple tests were conducted.

Testing Environment

The prototype was placed on a flat surface.

Objects were moved toward the sensors at different speeds and distances to simulate approaching vehicles.

Different scenarios were created:

- i. No object present
- ii. Slowly approaching object
- iii. Moderately approaching object
- iv. Rapidly approaching object
- v. Very close object

The LCD messages and alert responses were recorded.

4.5 Experimental Test Cases

Table 4.3: Test Case 1- Safe Condition

Parameter	Value
Distance	≥ 40 cm
Speed	Low
Risk level	Safe
LCD Output	ALL SAFE
Buzzer	OFF
LED	OFF

Observation:

The system correctly identified that no collision risk existed.

Table 4.4: Test Case 2 - Warning Condition

Parameter	Value
Distance	35 cm
Speed	Low

Risk level	Warning
LCD Output	WARNING
Buzzer	Slow Beep
LED	OFF

Observation:

The system detected a nearby object and generated an early warning.

Table 4.5: Test Case 3- High Risk Condition

Parameter	Value
Distance	22 cm
Speed	Moderate
Risk level	High Risk
LCD Output	HIGH RISK
Buzzer	Fast Double Beep
LED	OFF

Observation:

The system successfully increased the alert intensity as the object moved closer.

Table 4.6: Test Case 4- Critical Condition

Parameter	Value
Distance	8 cm
Speed	High
Risk Level	Critical
LCD Output	CRITICAL
Buzzer	Continuous
LED	Fast Blinking Red

Observation:

The system generated the maximum alert level to warn the rider of immediate danger.

4.6 Sample Results**Table 4.7: Left Side Direction**

Distance (cm)	LCD Output	Alert
55	ALL SAFE	None
38	WARNING	Slow Beep
24	HIGH RISK	Fast Beep
8	CRITICAL	Continuous Buzzer + LED

Table 4.8: Right Side Direction

Distance(cm)	LCD Output	Alert
50	ALL SAFE	None
36	WARNING	Slow Beep
22	HIGH RISK	Fast Beep
9	CRITICAL	Continuous Buzzer + LED

Table 4.9: Back side direction

Distance(cm)	LCD Output	Alert
48	ALL SAFE	None
39	WARNING	Slow Beep

23	HIGH RISK	Fast Beep
7	CRITICAL	Continuous Buzzer + LED

Table 4.10: Multi-Directional Results

Scenario	Left (cm)	Right (cm)	Back (cm)	LCD Output	Alert
Warning	39	35	50	WARNING L:39 R:35	Slow Beep
High Risk	29 but speed change > 5	22	45	HIGH RISK L:29 R:22	Fast Beep
Critical (Single)	14 but speed change >10	28	40	CRITICAL L: 14	Continuous Buzzer + LED
Critical (Multi)	55	22	9	CRITICAL B:9 R:22	Continuous Buzzer + LED

4.7 LCD Output Examples

The LCD display provides real-time information to the rider.

Safe State

- i. ALL SAFE

Warning State

- ii. WARNING
B: 36

High Risk State

- iii. HIGH RISK
R:22

Critical State

- iv. CRITICAL

B:8

Multi-Directional LCD Output

Case 1 - (Left+ Right + Back)

- i. Left: 35 cm (Warning)
- ii. Right: 27 cm but speed change > 5 cm/s (High Risk)
- iii Back: 18 cm (High Risk)

Final LCD: **HIGH RISK**

B:18cm R: 27cm

Case 2 (Left + Right)

- i. Left: 12 cm (Critical)
- ii. Right: 22 cm (High Risk)

Final LCD: **CRITICAL**

L:12cm R:22cm

Case 3 (Left + Right + Back)

- i. Left: 18 cm but speed change > 10 cm/s (Critical)
- ii. Right: 22 cm (High Risk)
- iii. Back: 9 cm (Critical)

Final LCD: **CRITICAL**

L:18cm R:22cm B:9cm

4.8 Results

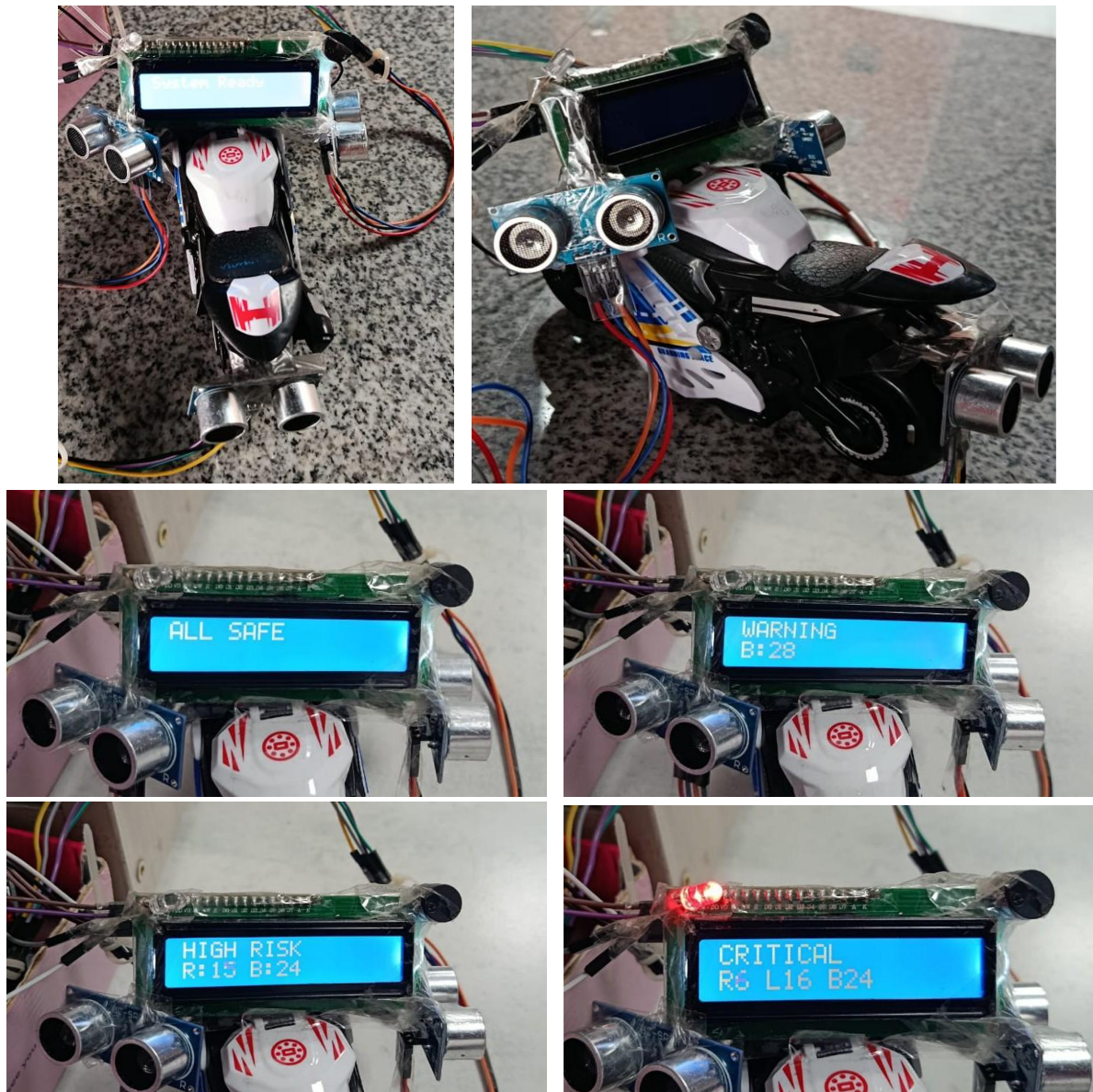
The experimental results indicate that:

- i. The system successfully detected objects from all three monitored directions.
- ii. Distance measurements remained stable due to averaging of sensor readings.
- iii. Speed approximation improved risk assessment compared to distance-only detection.
- iv. The LCD effectively displayed direction and distance information.
- v. The buzzer provided clear audible alerts.
- vi. The red LED effectively highlighted critical situations.

vii. The system responded in real time and successfully differentiated between Safe, Warning, High Risk, and Critical conditions.

The system evaluates all three directions independently and simultaneously, but the final alert displayed on the LCD is based on the highest risk level detected among them.

4.9 Prototype Demonstration



During demonstration, the following sequence was observed:

1. The system displayed System Ready after startup.
2. Objects approaching from the left, right, or back were detected.
3. The LCD displayed the corresponding direction and distance.
4. Alert intensity increased as the object approached.
5. Continuous buzzer and blinking red LED were activated in critical situations.
6. Once the object moved away, the system automatically returned to the safe state.

4.10 Implementation Outcome

The developed prototype successfully fulfilled all project objectives.

The system demonstrated:

- i. Multi-directional vehicle detection
- ii. Real-time distance measurement
- iii. Speed-based risk assessment
- iv. Intelligent alert generation
- v. Low-cost implementation
- vi. Practical applicability for bicycle and rickshaw safety

5. Analysis and Evaluation

5.1 Performance Analysis

The developed system was evaluated based on detection capability, responsiveness, reliability, cost-effectiveness, and usability. The prototype successfully detected approaching objects from the left, right, and back directions. Unlike traditional bicycle safety mechanisms that rely solely on rider observation, the proposed system continuously monitored surrounding areas and provided automatic alerts.

The use of three ultrasonic sensors ensured wider coverage of blind spots and improved rider awareness.

5.2 Accuracy Evaluation

The HC-SR04 ultrasonic sensors provided reasonably accurate distance measurements within the operating range.

To improve accuracy:

- i. Five consecutive readings were collected.

- ii. Average distance values were calculated.
- iii. Sudden fluctuations were reduced.

This averaging technique minimized measurement noise and reduced false detections.

The system produced stable distance readings during laboratory testing.

5.3 Risk Assessment Evaluation

One of the major strengths of the project is the integration of speed approximation with distance measurement.

Traditional obstacle detection systems often rely only on distance.

In contrast, the proposed system evaluates:

- i. Current distance
- ii. Change in distance over time

This allows the system to distinguish between:

- i. A stationary nearby object
- ii. A rapidly approaching vehicle

As a result, alerts become more meaningful and realistic.

5.4 Alert System Evaluation

The warning mechanism was evaluated under multiple test conditions.

Safe Condition

- i. No unnecessary alerts generated.
- ii. LCD displayed safe status.

Warning Condition

- i. Slow buzzer beep provided early awareness.

High Risk Condition

- i. Faster buzzer pattern indicated increased danger.

Critical Condition

- i. Continuous buzzer and blinking LED provided maximum warning.

This gradual escalation improves user understanding and reduces panic.

5.5 Cost Evaluation

The project was developed using low-cost components.

Table 5.1 : Cost of Components

Component	Approximate Cost (BDT)
Arduino UNO R3 (with USB cable)	650
HV-SR04 Ultrasonic Sensors (3 sensors)	310
16x2 LCD Display (12C Module)	250
Buzzer, Resistor (220 ohm) and Red LED	70
Wires and Breadboard	140
Toy Bicycle Model	380
18650 3.7v Lithium Battery with holder and switch	150
Total Cost	BDT 1950

Compared to commercial vehicle safety systems, the proposed solution is significantly more affordable.

5.6 Advantages

The developed system offers several advantages:

- i. Low-cost implementation.
- ii. Real-time operation.
- iii. Multi-directional monitoring.
- iv. Easy installation.
- v. User-friendly LCD display.
- vi. Intelligent risk evaluation.
- vii. Portable design.
- viii. Suitable for bicycles and rickshaws.

5.7 Limitations

Despite its successful implementation, several limitations were observed.

Environmental Sensitivity

Ultrasonic sensors may be affected by:

- i. Rain
- ii. Dust
- iii. Mud
- iv. Uneven road surfaces

Traffic Congestion

In highly crowded traffic conditions:

- i. Multiple vehicles may be detected simultaneously.
- ii. Occasional false alerts may occur.

Speed Estimation

The system uses an approximation technique rather than actual vehicle speed measurement.

Therefore:

- i. Speed calculations may not always represent exact vehicle speed.

Detection Range

The HC-SR04 sensor has a limited operating range and field of view.

5.8 Prevention

- i. The system can be manually switched off during heavy traffic conditions to reduce unnecessary alerts.
- ii. A waterproof ultrasonic sensor (such as JSN-SR04T) can be used instead of HC-SR04 for better outdoor performance.
- iii. Protective casing should be added to prevent rain and dust interference.
- iv. Sensor cleaning or covering mechanisms can improve reliability in real road environments.

5.9 Comparison with Existing Approach

Table 5.2: Comparison

Feature	Traditional Bicycle	Proposed System
Blind Spots Awareness	Limited	High
Real-Time Monitoring	No	Yes
Automatic Warning	No	Yes
Distance Measurement	No	Yes
Risk Analysis	No	Yes
Collision Alert	No	Yes
Cost	Low	Low

5.10 Future Improvements

Several improvements can be incorporated in future versions:

1. AI-based collision prediction.
2. Mobile application integration.
3. GPS tracking.
4. Camera-based object recognition.
5. Radar sensors for higher accuracy.
6. Waterproof enclosure.
7. Rechargeable battery management system.
8. Motorcycle and scooter adaptation.

5.11 Overall Evaluation

Based on testing and observations, the system successfully achieved its intended objectives.

The project demonstrates that a simple embedded system using low-cost sensors and an Arduino microcontroller can provide effective real-time collision awareness for bicycle and rickshaw riders.

The results suggest that the proposed system has strong potential for future real-world deployment after further refinement.

6. Summary

The Multi-Directional Bicycle Blind Spot & Collision Alert System was successfully designed and implemented as a low-cost embedded safety solution for bicycle and rickshaw riders. The system addresses a critical real-world problem where riders are unable to monitor vehicles approaching from blind spots, especially from the rear and sides.

The system uses three ultrasonic sensors to continuously monitor surrounding distances and an Arduino Uno to process the data in real time. Based on distance variation and speed approximation, the system intelligently classifies risk into Safe, Warning, High Risk, and Critical states.

A 16×2 LCD display provides real-time visual feedback, while a buzzer and red LED deliver audible and visual alerts during dangerous situations. The system ensures that alerts are only triggered when there is a genuine collision risk, thereby reducing false alarms.

Overall, the project demonstrates an effective combination of simplicity, affordability, and real-time responsiveness. It can significantly improve rider awareness and reduce accidents caused by blind spots in urban traffic environments.

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